Lab 3: Data-Centric Machine Learning with Scikit-Learn

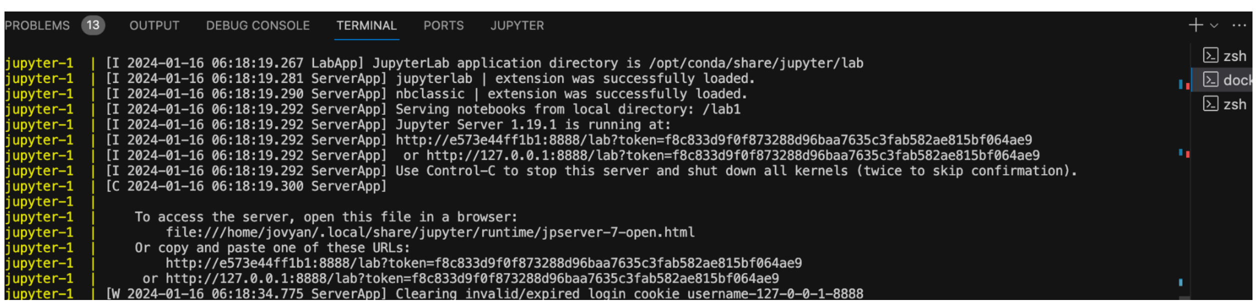
In this lab, you will be using Scikit-Learn to explore various aspects of data centric machine learning, including classification, clustering, dimensionality reduction, and regression. Below are the instructions to get started with the lab.

Setup Instructions Docker

We will be using a docker environment similar to the one which we used in lab 1. Download the lab folder on your local machine and from the lab directory, run this command (where your docker-compose.yml file is located):

```
docker-compose up --build
```

After running docker compose command you will see a url as shown in the image below. Paste the url in the web browser or paste it after selecting jupyter kernel in vscode to open the jupyter notebook.



There are two notebooks in which you have to work on and submit named,

1. lab3_regression.ipynb
2. lab3_classification.ipynb

There are two tutorial notebook as well which you can refer

1. tutorial_regression.ipynb
2. tutorial_classification.ipynb

Setup Instructions without Docker (Not Recommended)

If you are facing issues with docker installation or setup, then you can use any other environment setup which you already have like Conda or Python, you would have to install the necessary libraries which are imported in the notebook.